Pointed Lipschitz fractal interpolation on intervals and curves

FA/Banach run `fa_banach_001`, lane 1

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Source metadata and status

The source is P. R. Massopust, *Fractal Interpolation over Curves*, arXiv:2209.01033 (2022), published in *Contemporary Mathematics* 797 (2024), 61–73 [1]. Section 7, page 13 of the arXiv PDF, leaves two research directions to the reader. The second is to replace the unital Lipschitz algebra by the algebra of bounded real-valued Lipschitz functions vanishing at a base point.

Claimed status: full solution, likely valid. The theorem below carries out that pointed-Lipschitz direction for the interval partitions used in fractal interpolation, gives exact junction conditions, and transports the result to simple bi-Lipschitz curves in Banach spaces.

The source direction

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7. Future Research Directions

The above approach may be generalized along different directions.

- Instead of considering curves in Banach spaces, one may look in general at Banach or even Fréchet submanifolds.
- The Lipschitz algebras $\mathrm{Lip}(X, \mathsf{F})$ could be replaced by the space of bounded real-valued Lipschitz functions on X which vanish at a given base point $x_0 \in X$. The corresponding Lipschitz algebras are then denoted by $\mathrm{Lip}_0(X, \mathsf{F})$.

We leave it to the interested reader to pursue these research directions.

In the notation of the source, the direction is to replace $\mathrm{Lip}(X, F)$ by the pointed space

$$\mathrm{Lip}_0(X) = \{f : X \to \mathbb{R} : f \text{ is Lipschitz and } f(x_0) = 0\}.$$

On a bounded pointed metric space this is a Banach space for the norm $\|f\|_{\mathrm{Lip}_0} = \mathrm{Lip}(f)$; the base-point condition implies $\|f\|_\infty \le \mathrm{diam}(X)\,\mathrm{Lip}(f)$.

Proof intuition

The pointed condition changes the join-up problem. At an internal knot, the right branch sees $f(0) = 0$, whereas the left branch sees the unrestricted value $f(1)$. Consequently, an RB formula acts on *all* of Lip_0 precisely when the left vertical multiplier vanishes at that endpoint and the two affine terms agree there.

For contraction, multiplication contributes two terms: $\|s\|_\infty\,\mathrm{Lip}(f - g)$ and $\mathrm{Lip}(s)\|f - g\|_\infty$. The pointed estimate $\|f - g\|_\infty \le \mathrm{Lip}(f - g)$ combines them, while composition with an inverse partition branch contributes $\mathrm{Lip}(h_i^{-1})$. The junction conditions make the branchwise estimates glue without any additional loss on an interval.

Pointed fixed-point theorem

Fix a partition

$$0 = t_0 < t_1 < \cdots < t_n = 1.$$

For $1 \le i \le n$, let $h_i : [0,1] \to [t_{i-1}, t_i]$ be an increasing bi-Lipschitz homeomorphism, with $h_i(0) = t_{i-1}$ and $h_i(1) = t_i$. Put

$$\lambda_i = \mathrm{Lip}(h_i^{-1}), \qquad a_i = \|s_i\|_\infty, \qquad b_i = \mathrm{Lip}(s_i),$$

where $q_i, s_i \in \mathrm{Lip}([0,1])$.

For $f \in \mathrm{Lip}_0([0,1])$, consider the branch formula

$$(Tf)(x) = q_i(h_i^{-1}x) + s_i(h_i^{-1}x) f(h_i^{-1}x), \qquad x \in [t_{i-1}, t_i]. \tag{1}$$

Theorem 1. *The formula* (1) *defines an affine self-map of all of* $\mathrm{Lip}_0([0,1])$ *if and only if*

$$q_1(0) = 0, \qquad s_i(1) = 0, \quad q_i(1) = q_{i+1}(0) \qquad (1 \le i < n). \tag{2}$$

Under these conditions,

$$\mathrm{Lip}(Tf - Tg) \le c\, \mathrm{Lip}(f - g), \qquad c := \max_{1 \le i \le n} \lambda_i (a_i + b_i). \tag{3}$$

In particular, if $c < 1$, there is a unique $\psi \in \mathrm{Lip}_0([0,1])$ satisfying

$$\psi(h_i(t)) = q_i(t) + s_i(t)\psi(t), \qquad t \in [0,1], \quad 1 \le i \le n. \tag{4}$$

For every $f_0 \in \mathrm{Lip}_0([0,1])$, the iterates $f_k = T^k f_0$ converge to ψ in pointed Lipschitz norm and obey

$$\mathrm{Lip}(\psi - f_k) \le \frac{c^k}{1-c} \mathrm{Lip}(Tf_0 - f_0). \tag{5}$$

Proof. First suppose (1) defines a self-map on all of Lip_0. At $0 = h_1(0)$, taking $f = 0$ gives $(Tf)(0) = q_1(0)$, hence $q_1(0) = 0$. At the knot $t_i = h_i(1) = h_{i+1}(0)$, equality of the two branch values gives, for every $f \in \mathrm{Lip}_0$,

$$q_i(1) + s_i(1)f(1) = q_{i+1}(0) + s_{i+1}(0)f(0) = q_{i+1}(0).$$

Taking first $f = 0$ and then $f(t) = t$ proves the remaining conditions in (2).

Conversely, assume (2). It makes the two definitions in (1) agree at every internal knot, and it gives $(Tf)(0) = 0$. On the ith interval, composition and the product estimate show that

$$\mathrm{Lip}\big((Tf)|_{[t_{i-1},t_i]}\big) \le \lambda_i\big(\mathrm{Lip}(q_i) + a_i\,\mathrm{Lip}(f) + b_i\|f\|_\infty\big),$$

so Tf is continuous and finitely piecewise Lipschitz. Such a function on an interval is globally Lipschitz: telescope across the intervening knots and use that the subinterval lengths sum to the total length. Thus T maps Lip_0 to itself.

Let $u = f - g$. On the ith branch,

$$(Tf - Tg)(x) = (s_i u)(h_i^{-1}x),$$

and hence

$$\begin{aligned}\operatorname{Lip}\big((Tf - Tg)|_{[t_{i-1},t_i]}\big) &\le \lambda_i\big(a_i \operatorname{Lip}(u) + b_i\|u\|_\infty\big)\\ &\le \lambda_i(a_i + b_i)\operatorname{Lip}(u),\end{aligned}$$

because $u(0) = 0$ and the interval has diameter one. Moreover, at every internal knot the left difference is $s_i(1)u(1) = 0$ and the right difference is $s_{i+1}(0)u(0) = 0$. The same telescoping argument therefore gives the global estimate (3).

Finally, $\operatorname{Lip}_0([0,1])$ is complete for the norm $\operatorname{Lip}(\cdot)$. Indeed, a Cauchy sequence in that norm is uniformly Cauchy because every member vanishes at zero; its uniform limit is pointed Lipschitz, and passage to the limit in difference quotients gives convergence in Lipschitz norm. If $c < 1$, Banach's fixed-point theorem now gives the unique fixed point, convergence of the iterates, and (5). The fixed-point equation is exactly (4). □

Transfer to curves

Corollary 2. *Let E be a Banach space and let $\gamma : [0,1] \to E$ be a bi-Lipschitz homeomorphism onto its trace $X = \gamma([0,1])$. Set $x_0 = \gamma(0)$ and*

$$H_i = \gamma \circ h_i \circ \gamma^{-1}, \qquad Q_i = q_i \circ \gamma^{-1}, \qquad S_i = s_i \circ \gamma^{-1}.$$

If (2) and $c < 1$ hold, there is a unique $\Psi \in \operatorname{Lip}_0(X)$ such that

$$\Psi(H_i x) = Q_i(x) + S_i(x)\Psi(x), \qquad x \in X, \quad 1 \le i \le n.$$

It is $\Psi = \psi \circ \gamma^{-1}$, where ψ is the fixed point from Theorem 1. The transported RB iterates converge in $\operatorname{Lip}_0(X)$.

Proof. Composition with γ and γ^{-1} gives mutually inverse bounded linear maps between $\operatorname{Lip}_0(X)$ and $\operatorname{Lip}_0([0,1])$. Under these maps, the curve RB operator is conjugate to (1). Existence, uniqueness, and convergence therefore transfer directly. □

In particular, the corollary applies to every simple regular C^1 arc on a compact interval once it is given a bi-Lipschitz parametrization (for example, the usual regular embedded arcs used by interval-based fractal interpolation).

Why the contraction constant needs both terms

The two summands in (3) cannot generally be replaced by their maximum. Take $n = 1$, h_1 the identity, $q_1 = 0$, and $s_1(t) = at$ with $0 < a < 1$. This defines $Tf(t) = atf(t)$ on all of $\operatorname{Lip}_0([0,1])$. For $f(t) = t$,

$$\operatorname{Lip}(f) = 1, \qquad \operatorname{Lip}(Tf) = \operatorname{Lip}(at^2) = 2a.$$

Thus T is not a contraction whenever $a > 1/2$, even though both $\|s_1\|_\infty = a$ and $\operatorname{Lip}(s_1) = a$ are separately less than one. The coefficient in Theorem 1 is $2a$, and is attained on this test function.

Verification, scope, and novelty

The proof is exact and has no numerical or computer-assisted component. Its checks are: completeness of the pointed norm; necessity and sufficiency of the junction identities; the product Lipschitz estimate; inverse-branch expansion; finite interval gluing; and conjugacy under a bi-Lipschitz curve parametrization.

The result completes the source's pointed-Lipschitz direction for the natural interval partition model and its simple bi-Lipschitz curve transports. It does not address the source's separate, much broader direction concerning arbitrary Banach or Fréchet submanifolds.

The local run indexes were searched for arXiv:2209.01033, the title, pointed Lipschitz algebras, Lip_0, and RB operators. Bounded web searches on July 21, 2026 used the exact Section 7 wording and the combinations "pointed Lipschitz", "Lip_0", "fractal interpolation", and "Read–Bajraktarević". The arXiv record, 2024 publication metadata, and current author-publication search results were checked. No later work explicitly carrying out this direction was found. Novelty confidence is moderate because the fixed-point argument is elementary; proof confidence is high.

Human review recommendation. Review as a likely valid full answer to the second Section 7 direction. The most useful check is that "replace by Lip_0" is intended as a self-map on the whole pointed algebra; under that literal interpretation, the junction conditions in (2) are forced, not optional.

References

[1] P. R. Massopust, *Fractal Interpolation over Curves*, arXiv:2209.01033 (2022); *Contemporary Mathematics* 797 (2024), 61–73, doi:10.1090/conm/797/15934.